

Avni Badiwale

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Education

University of Wisconsin–Madison

Bachelor of Science in **Computer Science & Mathematics**

Relevant Coursework: Foundation Models, Probability Theory, Machine Organization, User Interfaces

Madison, Wisconsin

Sep 2024–May 2028

Work Experience

Garmin

Software Engineering Intern (Aviation Department)

Olathe, Kansas

May 2026–August 2026

- Developed features for a WPF/C# (MVVM) ground-support application that configures electronic circuit-breaker protection profiles and motor-control channels for a safety-critical avionics power distribution unit.
- After a misconfigured ECB blew up during testing, resolved the issue by building an interactive I²t trip-curve tool within the application that gives engineers live visual verification of overcurrent-protection profiles.
- Made the tool usable in real time by extending the third party library with custom adaptive sampling, adding logarithmic-axis support, and implementing geometric-error curve fitting to match the log-scale physics of I²t.
- Automated enforcement of the team's C# style conventions with a Roslyn script that rewrites member ordering across the codebase, eliminating hours of manual cleanup per cycle.

Epic Systems

Student Assistant

Verona, Wisconsin

May 2025–Jan 2026

- Proposed and delivered a campus-wide indoor webmap for 15,000 employees, replacing a legacy custom PDF setup that required manual updates for any changes for 37 buildings.
- Built the frontend in React with the ArcGIS SDK for JavaScript, rendering dynamic level-switching and spatial search across tens of thousands of indexed polygons.
- Developed RESTful APIs in C# to bridge two independent backends, unifying live spatial floor plan data with enterprise SQL employee records.
- Leveraged this unified data pipeline to enable real-time room availability, employee directory lookups, and inform interactive popups for every room within the webmap.
- Unblocked homepage integration by migrating a legacy Webpack 4 build to Vite/Esbuild, a prerequisite for supporting the ArcGIS SDK, and cut frontend build times by ~70% in the process.

Google Summer of Code at OpenMS

Software Engineering Intern

([Proposal](#) | [Project Summary](#)) Remote

May 2025–Sep 2025

- Integrated Apache Arrow and Parquet into the C++ build system of OpenMS, implementing columnar file I/O to significantly reduce data parsing overhead for downstream machine learning pipelines.
- Unblocked the entire Arrow integration on Ubuntu, macOS, and Windows by resolving incompatible Nlohmann/JSON versions independently vendored by Apache Arrow, OpenMS, and three host OSes.
- Eliminated the Windows onboarding barrier for OpenMS contributors by pioneering the project's first Conda-based build environment, reducing a multi-hour manual CMake setup to a single reproducible command.

Personal Projects

MergeNB (VSCode Extension for Jupyter Notebook Conflict Resolution)

[GitHub](#)

- First-in-market resolution engine that renders Jupyter Notebook git merge conflicts as interactive, cell-level rich diffs.
- Designed a cell-matching algorithm using Hungarian assignment and a weighted cost matrix to accurately align cells.
- Built a responsive UI using React and CodeMirror 6, powered by a custom local HTTP and WebSocket server for real-time state synchronization
- Added rich-output rendering via JupyterLab Rendermime, virtualized scrolling, and an auto-resolution engine.
- Engineered a 20-test CI/CD suite with GitHub Actions, utilizing Playwright browser automation to end-to-end test live Git merge-conflict resolution and UI rendering.

Technical Skills

Languages: C#, C++, Python, TypeScript/JavaScript, Java, SQL, Bash

Frontend & Web: React, WPF, Preact, HTML/CSS

Backend & Data: .NET, RESTful APIs, Apache Arrow, Parquet, PyTorch

Infrastructure: GitHub Actions (CI/CD), Docker, Linux, CMake, Conda, Playwright, Git